

Linalool's Multifaceted Antimicrobial Potential: Unveiling its Antimicrobial Efficacy and Immunomodulatory Role Against *Saprolegnia parasitica*

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Abstract

Saprolegnia parasitica (*S. parasitica*) is one of the most virulent oomycete species in freshwater aquatic environments, causing severe saprolegniasis and leading to significant economic losses in the aquaculture industry. Thus far, the prevention and control of saprolegniasis face a shortage of medications. Linalool, a natural antibiotic alternative found in various essential oils, exhibits promising antimicrobial activity against a wide range of pathogens. In this study, the specific role of linalool in protecting *S. parasitica* infection at both *in vitro* and *in vivo* levels was investigated. Linalool showed multifaceted anti-oomycetes potential by both of antimicrobial efficacy and immunomodulatory efficacy. For *in vitro* test, linalool exhibited strong anti-oomycetes activity and mode of action included: (1) Linalool disrupted the cell membrane of the mycelium, causing the intracellular components leak out; (2) Linalool prohibited ribosome function, thereby inhibiting protein synthesis and ultimately affecting mycelium growth. Surprisingly, meanwhile we found the potential immune protective mechanism of linalool in the *in vivo* test: (1) Linalool enhanced the complement and coagulation system which in turn activated host immune defense and lysate *S. parasitica* cells; (2) Linalool promoted wound healing, tissue repair, and phagocytosis to cope with *S. parasitica* infection; (3) Linalool positively modulated the immune response by increasing the abundance of beneficial Actinobacteriota; (4) Linalool stimulated the production of inflammatory cytokines and chemokines to lyse *S. parasitica* cells. In all, our findings showed that linalool possessed multifaceted anti-oomycetes potential which would be a promising natural antibiotic alternative to cope with *S. parasitica* infection in the aquaculture industry.

eLife assessment

This study presents **useful** findings on the efficacy and mechanism of linalool protection against *Saprolegnia parasitica* oomycetes in the grass carp model. The evidence is **incomplete** since the claims are partially justified, thus there is a need for more experimental data and more rigorous statistical data analysis. Revisions according to the recommendations will improve the work, making it of interest to scientists within the fields of aquaculture, ichthyology, microbiology, and drug discovery.

<https://doi.org/10.7554/eLife.100393.1.sa3>

1. Introduction

Saprolegniasis is one of the most serious oomycete diseases in the aquaculture industry all over the world, infecting a wide host range (fish, crustaceans, amphibians, and their eggs et al.) with a conspicuous symptom (grayish-white cotton wool-like covering on the surface of host)[1, 2]. Oomycete infections lead to significant economic losses and serious damage to natural ecosystems[1]. As the infection intensifies, saprolegniasis can cause muscle ulceration, compromised immune system, respiratory problems, secondary infection, and finally result in host death[3]. *Saprolegnia*, *Achlya*, and *Leptolegnia* are key pathogens responsible for Saprolegniasis outbreaks[4], among them, *Saprolegnia* is proven to be the most virulent[5, 6]. *S. parasitica*, *S. ferax*, *S. delica*, and *S. diclina* are the main pathogens causing saprolegniasis[7], as a highly virulent group belonging to the phylum Oomycota, *S. parasitica* serves as the primary species in fish infections[8].

In the past, malachite green was highly effective in controlling *saprolegnia* infections, however, it has been banned worldwide for its carcinogenic and toxicological effects, which led to a sharp re-emergence of *Saprolegnia* infections in aquaculture industry[9]. Recently, various methods have been established to deal with *S. parasitica* infections, mainly including physical (water quality management, temperature control, and so on), chemical (hydrogen peroxide, natural antimicrobial compounds, and so on), and biological (probiotics, antagonistic microbe, and so on) methods[10]. Among these, antibiotics have served as the primary therapeutic and prophylactic tools for Saprolegniasis management in aquaculture. However, their overuse and abuse have resulted in antibiotic resistance among various aquatic pathogens[11]. Therefore, the increasing need for sustainable and eco-friendly strategies to combat *saprolegnia* infections has driven research into natural antimicrobial agents, due to their advantages in terms of avoiding antibiotic resistance issues, health safety, and ecological sustainability[12].

In our earlier studies, the anti-oomycetes activities of 12 plant essential oils (EOs) and 5 of their major compounds (linalool, pinene, limonene, myrcene, and terpinene) were evaluated, and we found that linalool was one of the strongest anti-oomycetes compounds and selected for further study. Linalool, a monoterpene alcohol found in various EOs, has shown promising antimicrobial activity against a wide range of pathogens[13]. Our groups and other research teams have reported that linalool exhibits antimicrobial activity against bacteria such as *Escherichia coli*[12], *Listeria monocytogenes*[14], *Elizabethkingia miricola*, *Streptococcus pyogenes*[15], *Pseudomonas fragi*[16], *staphylococcus aureus*[17], *Brochothrix thermosphacta*[18] and *Aeromonas hydrophila*[19], as well as against oomycetes like *Colletotrichum lagenarium*[20].

and *Aeromonas hydrophila*[19], as well as against oomycetes like *Colletotrichum lagenarium*[20] and *Phytophthora capsici*[21], and fungi including *Aspergillus flavus*[22], *Penicillium citrinum*, *Chaetomium globosum*[23], *Colletotrichum gloeosporioides*, *Fusarium oxysporum*[24], *Botrytis cinerea*[25] and *Candida albicans*[26].

In our previous study, we found that linalool exhibited strong antimicrobial activity against *S. ferax*[27], while whether it is also a promising agent against *S. parasitica*, another key virulent pathogen for *Saprolegniasis* is still unknown, and the detailed mechanism, especially the *in vivo* immune mechanism induced by linalool remains poorly understood. Thus, in this study we focused on *S. parasitica* and aimed to investigate the specific role of linalool in protecting *S. parasitica* infection at both of *in vitro* and *in vivo* level by using different technologies, including *in situ* fluorescence observation, microstructure observation, Molecular Docking, AlphaFold2, transcriptome profiling, histological examinations, gut microbiota analysis, and deep bioinformatics analysis. We believe this study will shed light on the antimicrobial activity and mode of action of linalool against *S. parasitica*, and contribute to the development of novel antibiotic alternative strategies to cope with *S. parasitica* infection in aquaculture industry.

2. Results

2.1 The *in vitro* anti-oomycete activity of linalool against *S. parasitica* CQT2

Figure 1A illustrated the anti-oomycetes activities of EOs and their major components against *S. parasitica* CQT2. Notably, linalool and the EO extracted from daidai flowers were particularly effective in restraining mycelium growth.

Linalool exhibited the MIC and MFC of 0.025% and 0.025% (v/v, 0.216 mg/mL) against spores, respectively (**Figure 1B**). Besides, both the MIC and MFC against mycelium growth were 0.05% (v/v, 0.432 mg/mL) and 0.1% (v/v, 0.864 mg/mL), respectively. Various linalool doses were examined against *S. parasitica* CQT2 on linalool-containing plates to evaluate their impact on inhibiting mycelium growth. As shown in **Figure 1C-D**, the findings indicated the anti-oomycetes activity of linalool against *S. parasitica* CQT2 was dose-dependent. **Figure 1E** displayed mycelium growth on PDA plates under varying linalool concentrations. The viability of mycelium following linalool treatment was assessed using a PI staining assay, where non-viable (dead) mycelium appeared red after PI uptake. Untreated mycelium showed no detectable red fluorescent, indicating 100% viability. In contrast, distinct red fluorescent mycelium was observed after linalool treatment at both MIC and 2×MIC levels (**Figure 1F**).

2.2 Effects of linalool on cell membrane damage of *S. parasitica* CQT2

The *in situ* fluorescence observations were displayed in **Figure 2A**, with linalool treatment for 0 minutes, the mycelium structure remained undamaged, and the red and green fluorescence showed an even distribution. After 2 min, red fluorescence started appearing within the mycelium, accompanied by the entry of the styryl dye FM4-64, indicating damage to the cell membrane. As the duration of linalool treatment increased, the intensity of red fluorescence within the mycelium also increased, indicating a rise in the number of mycelial cell deaths along with severe damage to the cell membrane. When examined by SEM, the mycelium exhibited a smooth and intact surface in the control group (**Figure 2B**), while in the treated groups (**Figure 2C**), it appeared wrinkled, deformed, and shrunken (Red dashed quadrilateral). When examined by TEM, the untreated mycelium cell exhibited clear and intact ultrastructure in the cell membrane, cell wall, mitochondria, and rough endoplasmic reticulum (**Figure 2D**). However, after treatment with linalool, the cell membrane was wrinkled and seriously damaged, and

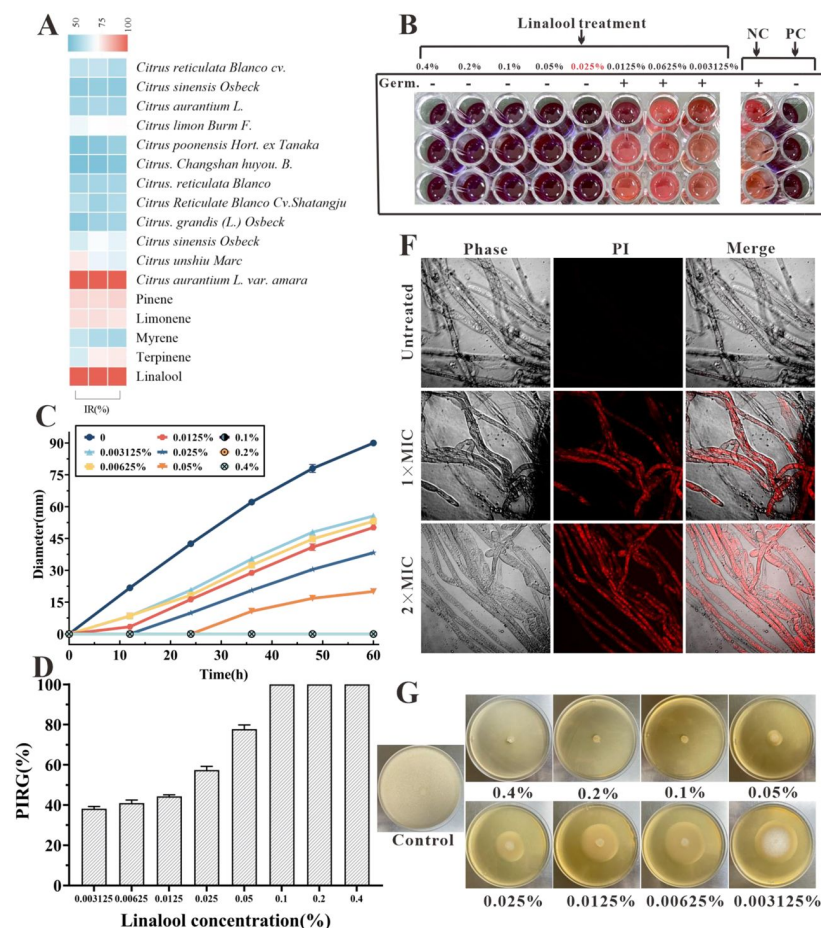


Figure 1.

The *in vitro* anti-oomycetes activity of linalool against *S. parasitica* CQT2. A) Inhibition rate (IR) of EOs and their major components against *S. parasitica* CQT2. B) Determination of MIC of linalool against *S. parasitica* spores, employing malachite green (2.5 mg/mL) as a positive control (PC) and Tween 20 as a negative control (NC). C) Effects of linalool on mycelium growth inhibition over a 6 h period at varying concentrations. D) Examination of PIRG % in mycelium treated with linalool after 60 hours. E) Effects of different concentrations of linalool on mycelium growth on PDA plates. F) Effects of linalool on the viability of *S. parasitica* CQT2 through a PI staining assay was conducted under 3 conditions: Control (no linalool), 1×MIC (0.05%) linalool treatment, and 2×MIC (0.1%) linalool treatment.

mitochondria began to dissolve (**Figure 2E**). As shown in **Figure 2F-G**, the cell membrane structure disappeared, the cell wall became thinner, and some exhibited severe deformation and bending. Vacuoles appeared in the cytoplasm, and mitochondria and rough endoplasmic reticulum ruptured and dissolved (red circles). No intact organelles were present within the cytoplasm.

As shown in the GO classification analysis (**Figure 2H**), the cellular components were distributed across 8 groups, with “cell part” and “membrane part” being the most prevalent. The DEPs related to intrinsic component of membrane were screened out, categorizing them into 4 classes: related to ribosome function, related to DNA/RNA function, related to signal transduction and regulation, and related to cell metabolism (**Figure 2I**). Therefore, the cell membrane may be a potential target for linalool.

2.3 The *in vitro* anti-oomycete mechanisms revealed by transcriptome analysis

As shown in Figure S1A-B, a total of 4142 differentially expressed genes (DEGs) were discovered between the control and treatment groups, with 2016 showing upregulation and 2108 downregulation. **Figure 2H** presented the Gene Ontology (GO) analysis, providing a comprehensive evaluation of the functional aspects of genes and their products. These DEGs were categorized into 20 GO categories, with 6 associated with biological activities, where cellular and metabolic functions had the most DEGs. There were also 6 molecular function categories, with catalytic activity and binding containing the majority of DEGs. Additionally, Figure S1C demonstrated the results of Kyoto Encyclopedia of Genes and Genomes (KEGG) annotation, revealing 10 metabolic (M) pathways, 2 environmental information processing (EIP) pathways, 2 cellular processes (CP), and 5 genetic information processing (GIP).

2.3.1 The ribosome biogenesis and protein synthesis of *S. parasitica* were prohibited by linalool

Figure 3A-B displayed the results of GO enrichment analysis, revealing the top 20 biological processes that differed significantly between the control and treated groups. Notably, in **Figure 3B**, the downregulated biological processes are highlighted, and remarkably, 19 of them were associated with “ribosome”, such as ribosome biogenesis, ribonucleoprotein complex biogenesis, pre ribosome, and small-subunit process some. In addition, KEGG enrichment analysis in **Figure 3C** revealed that “Ribosome biogenesis in eukaryotes” was the most enriched pathway, aligning with the GO enrichment analysis results mentioned earlier. The pathway of ribosome biogenesis in eukaryotes (**Figure 3D**) showed that linalool influenced DNA transcription, tRNA transport, rRNA processing and maturation (5.8S, 18S, and 25S), and the biogenesis and assembly of ribosome subunits (40S and 60S) in the cell, which might lead to the reduction of *S. parasitica* growth. We found that many DEGs related to ribosome biogenesis and RNA polymerase (**Figure 3E**) are down expressed. **Figure 3A** highlighted the upregulated biological processes, and notably, 11 of them were linked to “amino acids”, such as the sulfur amino acid, L-phenylalanine, alpha-amino acid, tyrosine, and homocysteine. Amino acids serve essential roles within the cell, acting as both the foundational components for synthesizing new proteins and as precursors for various metabolic processes[28].

Molecular docking was carried out to explore the specific binding protein of linalool with ribosome. The binding energy of linalool was -6.2 kcal/mol with NOP, -4.8 kcal/mol with SNU13, and -5.0 kcal/mol with DKC1 (Table S1). Consequently, NOP1 was selected as the receptor protein for subsequent molecular docking. **Figure 3F** illustrated the tertiary structure of ribosomal protein NOP1. Linalool formed hydrogen bonds with glutamic acid (Glu155) and phenylalanine (Phe156) of NOP1 while exhibiting hydrophobic interactions with alanine (Ala181), arginine

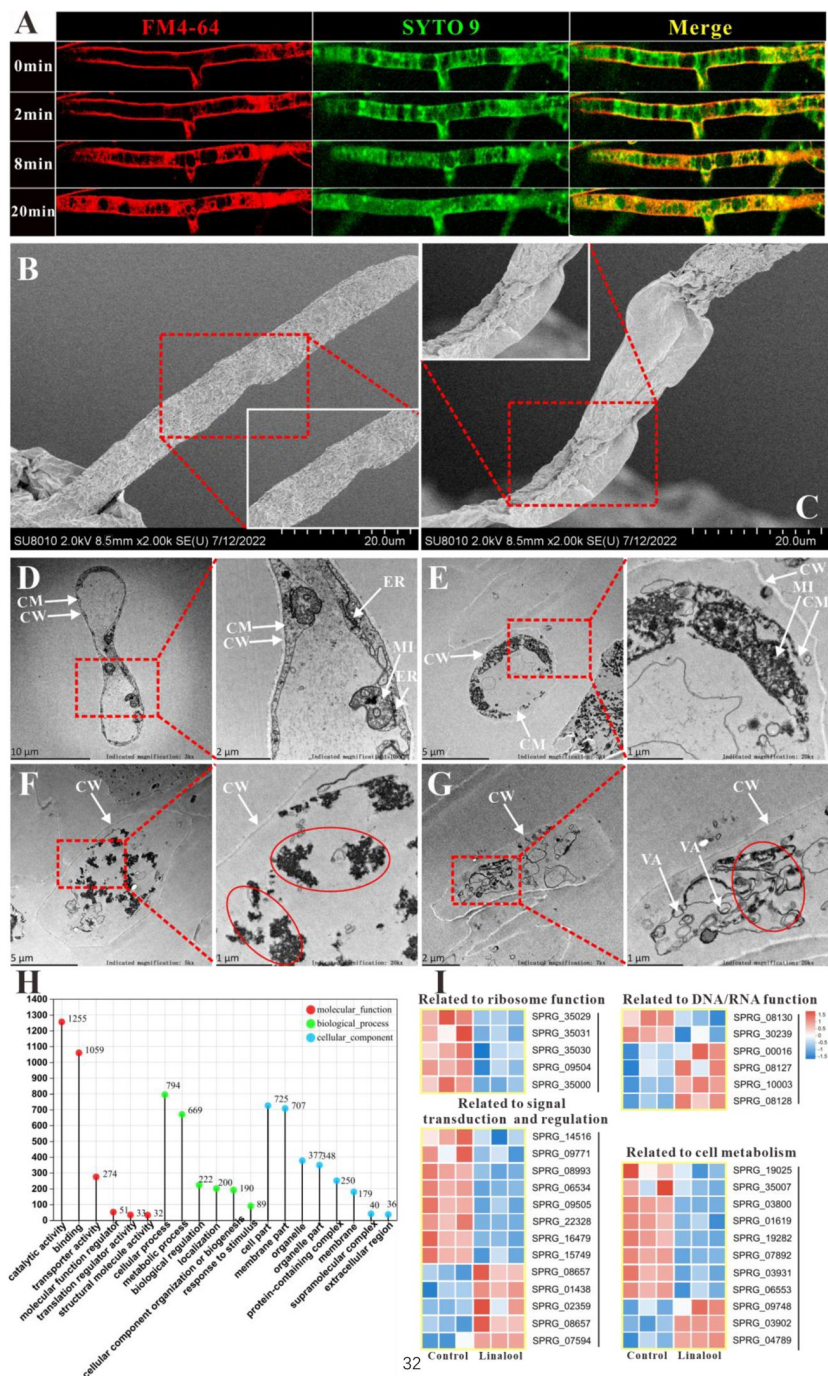


Figure 2.

Effects of linalool on the cell membrane integrity of *S. parasitica* CQT2. A) Observation of the effect of linalool on the cell membrane of *S. parasitica* CQT2 using the FM4-64 and SYTO 9 staining assay. B-C) SEM images of *S. parasitica* CQT2 mycelium without (B) and with (C) linalool treatment. D-G) TEM images of *S. parasitica* CQT2 mycelium without (D) and with (E-G) linalool treatment. CM: cell membrane; CW: cell wall; ER: rough endoplasmic reticulum; MI: mitochondria; VA: vacuoles. H) The GO classification of DEGs. I) DEPs involved in intrinsic component of membrane.

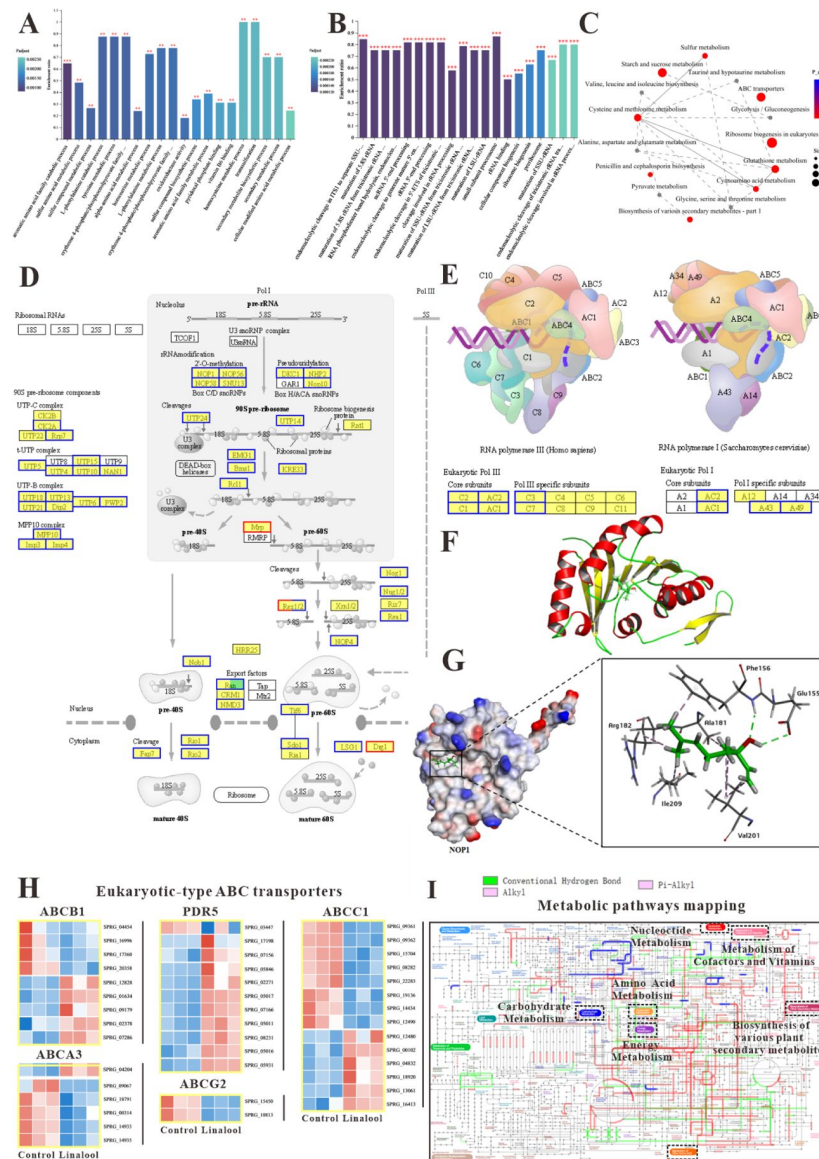


Figure 3.

The *in vitro* anti-oomycete mechanisms revealed by transcriptome analysis. A-B) The GO enrichment of up and down regulated DEGs. C) The KEGG enrichment of DEGs. Significant enrichment was labeled as “*”. $P < 0.05$, $P < 0.01$, and $P < 0.001$ were labeled as “**”, “***”, and “****”, respectively. D) Comparison of ribosome biogenesis in eukaryotes pathway between the linalool treated mycelium and the control group. E) Comparison of RNA polymerase pathway between the linalool treated mycelium and the control group. F) The tertiary structure of NOP1. G) Molecular docking of linalool with NOP1 involved enlarging the regions binding to the NOP1 activation pocket to showcase the detailed amino acid structures. H) Comparison of ABC transporters pathway between the linalool treated mycelium and the control group. The red squares represented up-regulated genes, and the blue squares represented down-regulated genes. I) Visual analysis of metabolic pathways with iPath3.0. The figure represented gene set annotated pathways, red and green represented pathways annotated by genes in different gene sets, respectively, and blue represented pathways co-annotated by genes in two gene sets.

(Arg182), valine (Val201), and isoleucine (Ile209) (**Figure 3G** [↗](#)). In summary, this interference may hinder the ribosome's ability to execute normal biological processes, such as protein synthesis, thereby impacting the growth and reproduction of oomycetes.

2.3.2 The global metabolisms of *S. parasitica* were regulated by linalool

In KEGG enrichment analysis, we found that 6 processes were associated with metabolic pathways, such as cysteine and methionine metabolism, starch and sucrose metabolism, and sulfur metabolism. Moreover, metabolic pathways were the most annotated pathway which was in line with the results of KEGG annotation analysis. As many changes were linked to metabolism, the Interactive Pathways Explorer (iPath) was employed to comprehend the global differential metabolic response in *S. parasitica* CQT2 (**Figure 3I** [↗](#)) which identified that linalool affects the global metabolic regulation of *S. parasitica* CQT2.

2.3.3 ABC transporters of *S. parasitica* were affected by linalool

Some other important pathways were significantly enriched in our study, such as ABC transporter and sulfur relay system. The DEGs involved in ABC transporters (ABCA3, ABCB1, ABCC1, ABCG2, and PDR5) were shown in **Figure 3H** [↗](#). In brief, these results demonstrated that cell membrane, protein synthesis, functions related to metabolism, and ribosome were significantly impacted by linalool and were potential drug targets of linalool.

2.4 The *in vivo* protective effect of linalool against *Saprolegniosis*

To study the protective effects of linalool on *S. parasitica* infection in fish, linalool was added to grass carp after *S. parasitica* infection (linalool therapeutic group, LT) or 48 h before infection (linalool prophylactic group, LP), and a positive control group (PC) without linalool and a negative control group (NC) without infection and linalool were set up (**Figure 4A** [↗](#)). The results showed that the survival rate of *S. parasitica* infected grass carp was significantly improved after treatment with linalool (**Figure 4C** [↗](#)), which proved greatly anti-*Saprolegnia* activity of linalool. In **Figure 4B** [↗](#), there was no observable mycelium in both LT and LP groups, while the PC group displayed significant mycelium growth along with inflammation on the dorsum. The LZM activity was significantly increased in LP and LT group on the seventh day (**Figure 4E** [↗](#)), yet the AKP activity (**Figure 4D** [↗](#)) was reduced. However, there was no significant difference in the SOD activity (**Figure 4F** [↗](#)) between LP and LT groups during the whole feeding trial.

To explore the histomorphology alterations caused by *S. parasitica* infection and linalool treatment, we performed histomorphology evaluations through HE staining (**Figure 4G** [↗](#)). The findings revealed significant pathological changes in the kidney, liver, spleen, and skin of grass carp due to *S. parasitica* infection. In PC group, severe kidney damage was evident, characterized by enlarged glomeruli, glomerular cysts, interstitial bleeding, and infiltration of inflammatory cells. In addition, extensive vacuolation and expansion of intercellular space was observed in hepatocytes, with nuclei displaced toward one side. Cytoplasmic pyknosis and structural disintegration were observed in the pancreatic acinar cells. Meanwhile, the red and white pulp of the spleen was poorly demarcated, the splenic corpuscle was irregularly shaped and arranged, and the white pulp was structurally disrupted. Furthermore, there was a notable increase in the number of macrophages and an expansion in the size of melanoma-macrophage centers. *S. parasitica* not only caused damage to both the dermal and epidermal layers but also induced muscle cell degeneration. Continuous treatment of linalool for 7 days significantly improved kidney, liver, spleen, and skin morphological integrity.

In the LP and LT groups, linalool alleviated tissue damage caused by *S. parasitica* infection on the dorsal surface of grass carp, enhancing the healing capacity (**Figure 4B** [↗](#)). Histomorphology demonstrated that linalool could improve the integrity of organ morphology in grass carp (**Figure**

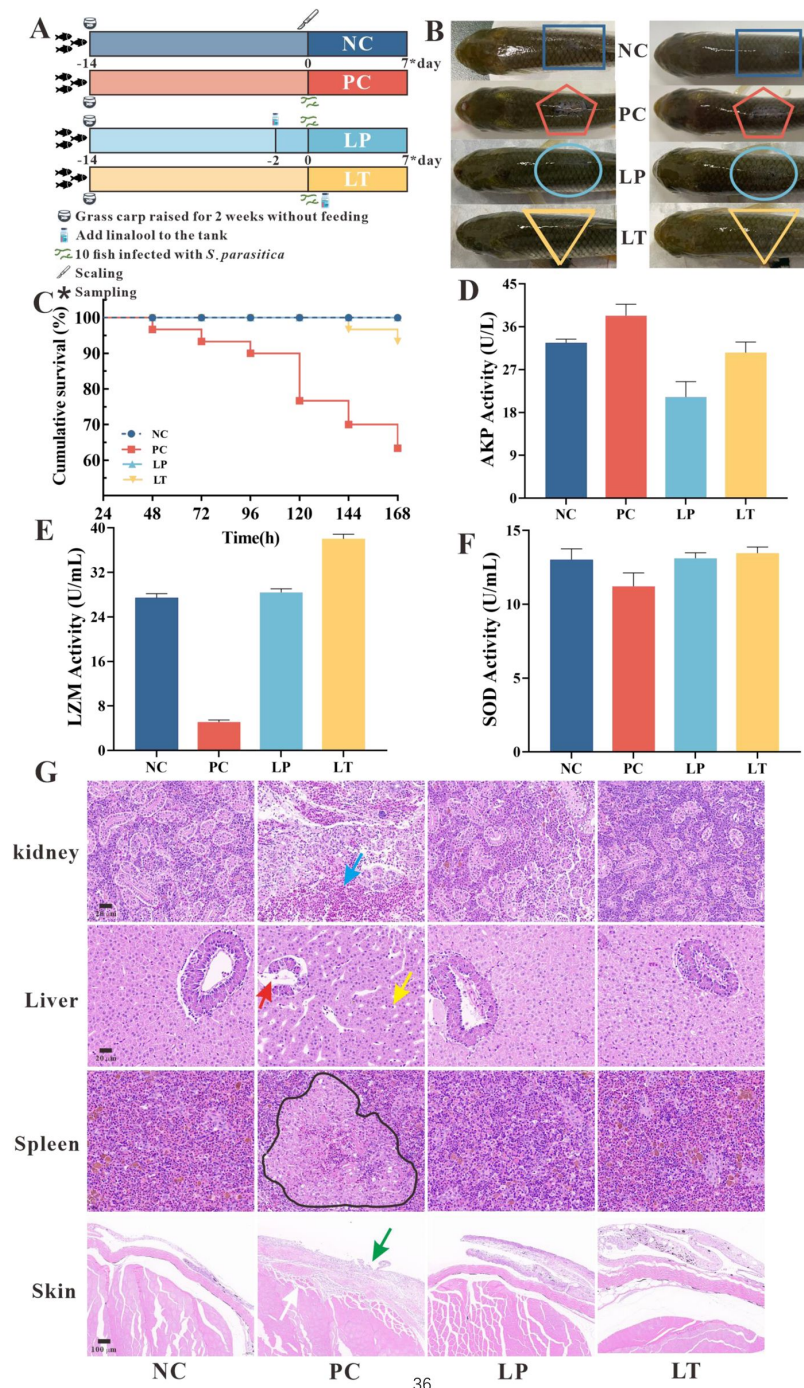


Figure 4.

Protective effect of linalool on grass carp infected with *S. parasitica*. A) The experimental design. Grass carp were raised for 2 weeks without feeding, fish without infection and linalool (Group NC), fish infected with *S. parasitica* (Group PC), and 10 fish uninfected soaked water containing linalool for 2 days and then 1×10^6 spores/mL secondary zoospores were added (Group LP), and fish infected with *S. parasitica* soaked for 7 days in water containing linalool (Group LT). B) The symptoms of *S. parasitica* infection in grass carp of different groups. C) The survival rates of grass carp infected with *S. parasitica* of different groups. D-F) Alkaline phosphatase (AKP), lysozyme (LYZ), and superoxide dismutase (SOD) activities in serum of grass carp of different groups. G) Histopathological analysis grass carp tissues in different groups. The arrows of different colors indicated: inflammatory cell infiltration in the kidney (blue arrow), cytoplasmic pyknosis (red arrow), nuclei displaced toward one side (yellow arrow), the red and white pulp was poorly demarcated, and a larger volume of melano-macrophage centers (black area), critical damage to the epithelium (green arrow) and myofiber (white arrow).

4G [↗](#)). Transcriptomics of the spleen revealed that linalool promoted wound healing, tissue repair, and phagocytosis to cope with *S. parasitica* infection (**Figure 5E** [↗](#)). In all, our results indicated that linalool promoted wound healing in grass carp.

2.5 The *in vivo* Saprolegniasis protective mechanism of linalool revealed by comparative transcriptome analysis

To analyze the contrasting gene-level responses in grass carp spleens, a vital immune organ, following *S. parasitica* infection and linalool treatment, spleen samples were collected for transcriptome analysis. Compared to PC group, 2029 genes in LT group exhibited significant difference, with 809 DEGs up regulated and 1220 down regulated. Meanwhile, 671 genes were identified as significantly different, with 609 DEGs up-regulated and 618 downregulated in LP group relative to PC group (Figure S2A-B).

Based on the GO annotation analysis, a total of 2219 unigenes in the LT group and 520 unigenes in the LP group were classified into 3 primary functional categories (Figure S2C). Within the biological process category, “cellular process,” “metabolic process,” and “biological regulation” were the most prominent subcategories. In terms of cellular components, “membrane part” and “cell part” were highly represented. “Binding” and “catalytic activity” were the most prevalent subcategories within molecular functions. Regarding KEGG annotations, the largest cluster within organismal systems was the immune system, encompassing 209 genes (42.47%) in the LT group (Figure S2D) and 38 genes (50.00%) in the LP group (Figure S2E). In the environmental information processing category, most unigenes were associated with signal transduction, with 227 (65.60%) in the LT group and 38 genes (52.00%) in the LP group. These results suggested that the immune response and signal transduction pathways were crucial for grass carp in dealing with *S. parasitica* infections.

The GO enrichment analysis of the spleens illustrated in **Figure 5A-B** [↗](#) to reveal distinct biological process. Those down regulated biological process were shown in **Figure 5A** [↗](#), compared to PC group, the “positive regulation of immune system process” term (GO:0002684) and “regulation of body fluid levels” term (GO:0050878) were enriched with the greatest number of DEGs in LT group. It is worth noting that there are 9 terms related to “body fluids (especially blood)”, such as regulation of coagulation (GO:0050818), blood coagulation (GO:0007596), and regulation of hemostasis (GO:1900046). In addition, those up regulated biological process, the “ion binding” term (GO:0043167) was enriched with the greatest number of DEGs in LT group. **Figure 5B** [↗](#) illustrated the GO terms with the largest number of down regulated DEGs in LP group was “small molecule binding” (GO:0036094).

KEGG enrichment analysis was used to explore the functional pathways involving DEGs. The 20 significantly enriched up regulated and down regulated pathways of each group are shown in **Figure 5C-D** [↗](#). Multiple pathways associated with immune response were enriched in down regulated genes after linalool treatment in LT group, including complement and coagulation cascades, Toll-like receptor signaling pathway, IL-17 signaling pathway, C-type lectin receptor signaling pathway, T cell receptor signaling pathway, B cell receptor signaling pathway, and so on (**Figure 5C** [↗](#)). Toll-like receptor signaling pathway, RIG-I-like receptor signaling pathway, Complement and coagulation cascades, and Chemokine signaling pathway were enriched with mostly up-regulated genes. Furthermore, metabolic pathways were enriched with mostly down-regulated genes in LP group. Some pathways associated with immune response were significantly enriched, such as Antigen processing and presentation, Fc epsilon RI signaling pathway, Intestinal immune network for IgA production, and so on (**Figure 5D** [↗](#)). GO and KEGG enrichment analyses revealed the presence of numerous immune-related genes within corresponding functional processes and pathways. The expression of *fb*, *c1q*, *c2*, *c3*, *c4*, *c6*, *c7*, *c8A*, *c8b*, and *c9* genes were up expressed. Additionally, there was a significant up expressed in the gene expression levels of

coagulation factors II (thrombin), VII, VIIi, VIII, X, coagulation factor IXb and IXa, serpin peptidase inhibitor clade C/D/G/F, *fga*, *fgb*, *fgg*, protein C, plasminogen, and plasminogen activator (**Figure 5E** [↗](#)). The differential expression of immune-related DEGs implied their role in host defense against *S. parasitica* infection.

To ensure the regulation of immune-related DEGs by linalool, the expression of several vital immune related genes (*cfh*, *masp2*, *c1r*, *c3*, *cfb*, *ap-1*, *il-1 β* , and *il-6*) in different groups were tested by RT-qPCR assay. The results showed that the expression of these genes in the LP and LT groups significantly exceeded those in the PC group ($P < 0.05$) (**Figure 5H** [↗](#)). Further exploration of immune-related genes indicated that linalool treatment promoted the expression of genes linked to the complement system and inflammatory factors. Additionally, analysis of 8 genes revealed that the real-time fluorescence quantitative analysis results were consistent with the expression trend of transcriptome analysis genes.

2.6 Linalool regulated gut microbiota composition

In our study, 862 OTUs were obtained from 12 samples through 16S sequencing analysis, with 48 common OTUs shared across all samples (**Figure 6E** [↗](#)). Alpha diversity analysis (**Figure 6A-D** [↗](#)) showed that the Simpson, Shannon, Chao1, and Ace index of gut microbiota decreased after linalool treatment which indicated that linalool reduced the abundance and diversity of gut microbes. PCoA analysis showed a significant difference in microbial composition among the PC, LP, and LT groups (**Figure 6F** [↗](#)).

The composition of gut microbiota in **Figure 6G** [↗](#) showed that the main dominant bacteria in the intestinal tract are Proteobacteria, Actinobacteriota, Verrucomicrobiota, Firmicutes, and Bacteroidota at the phylum level. Among them, compared with PC group, an obvious tendency of greater relative abundances of phyla Proteobacteria and Actinobacteriota was evidenced in LP and LT groups, the relative abundance of Verrucomicrobiota was decreased. From the composition of the genus level, the *Neochlamydia* decreased, while *Achromobacter* increased. It is worth noting that the genus *Aurantimicrobium* (classified as Actinobacteriota) was increased in linalool prophylactic and linalool therapeutic groups. Notably, among biomarkers with an LDA SCORE exceeding 3, the genus *Actinomycetales* bacterium held the highest number of positions, which contributed to the major differences between the positive control and linalool treatment groups. Moreover, the potential metabolic pathways of gut microbiota primarily showed enrichment in functions related to metabolism, including carbohydrate, amino acid, and cofactors and vitamins metabolism (**Figure 6I** [↗](#)).

2.7 Correlation analysis of potential related immune genes, specific microbial species, and physical characteristics

Correlation analysis showed that complement system were significantly positively correlated to Actinobacteriota, AKP, and SR (survival rate), which, in turn, a significantly negatively correlated with Verrucomicrobiota. Additionally, there was a significant positive correlation between PPRs and AKP. Notably, IgA related immune genes were correlated significantly with Proteobacteria (**Figure 6J** [↗](#)). In summary, our findings suggested that linalool effectively modulated gut microbiota during *S. parasitica* infection which was associated with immune response.

3. Discussion

3.1 *In vitro* mode of action of linalool against *S. parasitica*

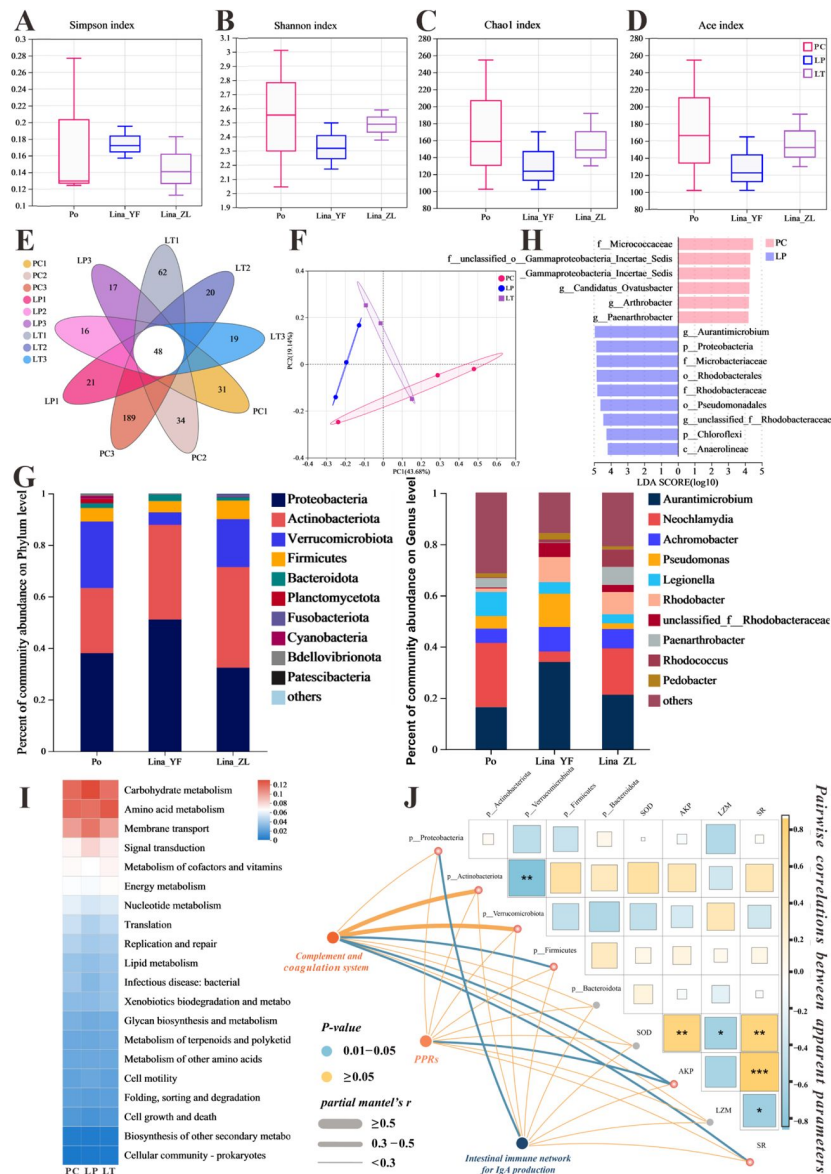


Figure 6.

The effect of linalool on regulating gut microbiota of grass carp infected with *S. parasitica* and correlation analysis. A-D) The α diversity index comparison among the different groups. E) The OTUs petal map of PC, LP, and LT groups. F) PCoA using Bray-Curtis distance revealed variations among the PC, LP, and LT groups (ANOSIM $R = 0.3086$, $P = 0.061$). G) Relative abundance of the top 10 species in the gut from the different groups (phylum and genus levels). H) Column chart of LDA value distribution. Discriminative biomarkers identified by linear discriminant analysis effect size (LEfSe) with logarithmic LDA score greater than 3.0. I) Heat map of the differences in predicted functional metabolisms within gut bacterial KEGG pathways. J) The correlation layout of potential related immune genes, specific microbial species, and physical characteristics. Pairwise comparisons of characterizations are presented, with a color gradient and block size denoting Pearson's correlation coefficients.

3.1.1 Membrane disruption

The cell membrane maintains cell integrity, regulates substance entry and exit, while the cell wall provides structural support, protection, and promotes growth and cell-to-cell connections[29]. The mycelium forms a dense structure to protect *S. parasitica* spores from external factors. CLSM, SEM, and TEM observations suggested that the cell membrane of the mycelium treated with linalool was damaged, causing the intracellular components to leak out, and could not develop completely (Figure 2A-G). Meanwhile, 678 genes related to the “intrinsic component of membrane” showed significant expressed in GO enrichment analysis, suggesting that linalool may enhance the expression of intracellular membrane components, reflecting its impact on cell membrane structure or function. Gao et al. illustrated that linalool caused significant damage to the cell wall/membrane of *E. coli*, resulting in the lysis and death of bacterial cells[12]. Gao et al. demonstrated that linalool[14], Changshan-huyou Y.B. Chang EO[30], and Fingered Citron EO[31] can disrupt the cell membrane of *L. monocytogenes*. Lemon essential oil nanoemulsions altered the permeability of *E. coli* cell membrane, affecting membrane potential, integrity, and efflux function[32]. Cruz et al suggested that alkamides in *Echinacea* disrupted the cell wall/membrane complex, making it an ideal target for specifically inhibiting pathogens[33]. Hinokitiol caused mycelium malformation by inducing cell component leakage and influencing chitinase activity[34]. Therefore, linalool may act anti-oomycete effects against *S. parasitica* by targeting the cell membrane.

ABC transporters utilize ATP hydrolysis energy to facilitate the export of harmful compounds or essential lipids across bacterial and eukaryotic membranes[35]. The DEGs involved in ABC transporters (ABCA3, ABCB1, ABCC1, ABCG2, and PDR5) in our study, almost all of which have been linked to illness states usually involving loss of function[36]. PDR5-related proteins in fungi played a role in excreting xenobiotics, including antifungal agents[37]. Multidrug resistance-associated protein (MRP) transporters were typically associated with multidrug resistance, cellular detoxification, and drug metabolism[38]. To sum up, linalool may greatly inhibit mycelium growth by interfering with protein synthesis from the “inside” and damaging the integrity and permeability of the cell membrane and wall from the “outside.”

3.1.2 Reduction of ribosome biogenesis and RNA polymerase

To further investigate the mechanism of linalool against *S. parasitica*, the gene expression profiles of *S. parasitica* were compared (with or without exposed to linalool). Many DEGs related to ribosome biogenesis and RNA polymerase were down expressed, indicating that the linalool may influence translation, replication, and repair processes, leading to increased protein diversity and potential disruption of corresponding functions[39]. Linalool influenced DNA transcription, tRNA transport, rRNA processing and maturation (5.8S, 18S, and 25S), and the biogenesis and assembly of ribosome subunits (40S and 60S) in the cell, which might lead to the reduction of *S. parasitica* growth. Ribosomes are assembled logistically by more than 200 ribosomal proteins (such as NOP, RIOK, and UTP) and RNA factors in a highly coordinated manner[40]. Depletion of Nop-7-associated 2 (NSA2) led to decreased nascent rRNA synthesis, reduced protein synthesis, and impacted cell proliferation due to its critical role in ribosome biogenesis regulation[41]. The inhibitory effect of myriocin on *Fusarium oxysporum* includes both membrane damage and interactions with intracellular targets, such as proteins like RIOK2, within the ribosome biogenesis pathway[42]. Innate immunity in *Caenorhabditis elegans* has been discovered to be negatively regulated by NOL-6 (also named UTP22)[43]. The growth of *Aspergillus niger* may be inhibited by cyclosporin A because it may prevent the transcription and translation of the RNA polymerase III complex[44]. Rifampicin can prevent RNA transcription in bacteria by interacting with the β -subunit of DNA-dependent RNA polymerase, which can then inhibit protein biosynthesis[45]. These results indicated that the down-regulation genes may limit the proteins necessary for normal cell biogenesis and growth. Therefore, ribosome is a promising potential target of linalool against *S. parasitica*.

3.1.3 The influence on the metabolism function of *S. parasitica*

Metabolism function is crucial for cells to maintain homeostasis, survive, and proliferate[46]. Linalool caused the activation of energy metabolism pathways, compensating for the insufficient energy needed for cell survival[47]. In our study, linalool was found to enhance amino acid metabolism, carbohydrate metabolism, energy metabolism, and so on in mycelial cells, which suggested that cells may boost functions related to energy metabolism to acquire additional energy for repairing damage caused by linalool. Hence, linalool treatment caused intracellular energy metabolism disruption, resulting in impediments to protein synthesis, cellular signal transduction, DNA synthesis, and ultimately affecting cell viability and proliferation.

3.1.4 The difference *in vitro* mechanisms between *S. parasitica* and *S. ferax*

In our previous study, we found that linalool affected *S. ferax* mainly by altering membrane permeability, causing permanent damage, and membrane breakdown; affecting the oxidation and metabolism of fatty acids and lead to mitochondrial dysfunction and the inhibition of energy supply; degrading the branched chain amino acids and result in the dysfunction of protein synthesis. While, for *S. parasitica*, linalool exhibited antimicrobial activity by prohibiting ribosome function, inhibiting protein synthesis, disrupting the cell membrane of the mycelium. We found several interesting phenomena when compared these two species. Firstly, the inhibition of protein synthesis was found in both species, but the interesting difference is that linalool inhibits protein synthesis through degrading the branched chain amino acids in *S. ferax*, while through prohibited ribosome function in *S. parasitica*. Secondly, linalool affected the oxidation and metabolism of fatty acids and mitochondrial dysfunction in *S. ferax*, while not found in *S. parasitica*. These phenomena showing above are very interesting results and we are eager to explore the biological mechanisms in the future.

3.2 *In vivo* mode of action of linalool against *S. parasitica* infection

Recent studies demonstrated that the spleen serves antibacterial functions in various fish species, with a high quantity of melano-macrophages that assist in phagocytosis during the immune response[48]. The rise of macrophages and the change in the pathological and physiological circumstances in fish are correlated[49]. Similarly, histopathological analysis demonstrated that linalool effectively reduced *S. parasitica* infection-induced damage in the spleen, indicating its potential immunomodulatory effects on immune organs. Here we used comparative transcriptome analysis to investigate potential genes and signaling pathways regulated by linalool in the spleen of grass carp. Several genes associated with both innate and adaptive immunity, such as complement molecules, pattern recognition receptors, inflammatory cytokines, immunoglobulins, and chemokines, were differentially expressed as shown in **Figure 5E-G**.

3.2.1 Enhancement of complement and coagulation system

It is widely acknowledged that the complement and coagulation system in fish is an important immune defense mechanism for fighting pathogens and maintaining immune homeostasis[50]. The complement system is made up of over 30 different proteins (complement receptors, internal components, and regulatory proteins) and 3 activation pathways (the classic pathway, the lectin pathway, and the alternative pathway)[51]. In our study, the expression of *fb*, *c1q*, *c2*, *c3*, *c4*, *c6*, *c7*, *c8a*, *c8b*, and *c9* genes were all up regulated after linalool therapeutic. Meanwhile, the *c3a* anaphylatoxin chemotactic receptor (*c3aar1*) and mannan-binding lectin serine protease 2 (*masp2*) genes were up expressed. During complement activation, *c3* and *c5* split into molecules such as *c3a*, *c3b*, *c5a*, and *c5b*, of which *c3a* and *c5a* are considered strong inflammatory mediators because of their ability to attract leukocytes and promote inflammation and pathogen clearance[52]. *masp2* activates complement components such as *c4* and *c2*, triggering a complement cascade reaction that ultimately leads to pathogen rupture[53]. *c8b* is a subunit of

the complement component 8 (*c8*) protein, which helps form the membrane attack complex (MAC) for cell lysis and immune defense against microbes[54]. In addition, high expression of complement factor H like 3 (*chf3*) prevents overactivation of the complement system to avoid damage to their own tissues[55]. The effective antimicrobial role of the *c5a-c5ar1* signaling pathway in the complement system has been identified. It functions by enhancing neutrophil phagocytosis of fungi and macrophage-mediated fungicidal activity to combat fungal infections[56]. In all, our findings suggested that linalool may enhance the complement system which in turn activated host immune defense and lysate *S. parasitica* cells.

The coagulation cascades are connected with the complement system. Linalool significantly increased the expression of fibrinogen alpha (*fga*), beta (*fgb*), and gamma (*fgg*) chain, which together encoded fibrinogen[57], fibrinogen is a crucial component of blood clots to prevent blood loss, and also important in the initial stages of wound healing[58]. Secondly, activated prothrombin *f2* is central in clot formation, platelet aggregation, and activating other clotting factors. It also supports tissue repair, cell growth, angiogenesis, and vascular integrity[54]. Thirdly, an increase in plasminogen (PLG) levels can cause an immune response[59], and serve as a regulator of the natural immune system by promoting the phagocytosis of phagocytes[60]. Taken together, the upregulation of fibrinogen, prothrombin, and plasminogen in the coagulation system suggested that linalool may promote wound healing, tissue repair, and phagocytosis to cope with *S. parasitica* infection.

3.2.2 Regulation of gut microbiota and increase the abundance of Actinobacteriota

Gut microbiota controls host growth and health status through various pathways[61]. Firstly, in our study, the function of gut microbiota is mainly enriched in metabolic-related pathways linalool increased the abundance of the beneficial phylum Actinobacteriota and decreased the abundance of the harmful phylum Verrucomicrobiota. Actinobacteriota plays a crucial role because Actinobacteria can produce enzymes, antibiotics, Short-chain fatty acids (SCFAs), signaling molecules, and immunomodulators[62]. Meanwhile, Actinobacteriota correlated positive with the complement and coagulation system according to the correlation analysis. Secondly, intestinal IgA production was correlated significantly with Proteobacteria which are the primary inducers of IgA production by B cells[63]. In summary, our findings suggested that linalool may positively modulate the immune response by increasing the abundance of beneficial phylum Actinobacteriota to *S. parasitica* infection.

3.2.3 Regulation of Inflammatory Factors

Pattern Recognition Receptors (PPRs) detect and initiate multiple signaling pathways upon recognizing Pathogen-Associated Molecular Patterns (PAMPs) on the surface of pathogens (bacteria, viruses, fungi, etc.), inducing subsequent host immune responses[64]. Myeloid differentiation primary-response protein 88 (MyD88) serves as the crucial signaling adaptor protein for Toll-like and interleukin-1 receptors, which is responsible for the conduction of multiple downstream signaling pathways[65]. The expression of *ap-1*, *p-38*, and *pl3k* were up regulate in downstream signaling pathways, causing higher expression of inflammatory factors, and enhanced cellular oxidation and antioxidant levels[66]. Meanwhile, in our study, the gene expression levels of *il-1 β* , *il-6*, *ccl19*, and *ccl5* which are closely related to the inflammasome, were up expression. Inflammasomes are protein complexes capable of detecting diverse inflammatory stimuli and are vital for eliminating pathogens or injured cells[67]. Therefore, linalool might have influenced the immune system response in grass carp, regulating the inflammatory processes.

4. Conclusions

In summary, our study proved that linalool is an effective natural antimicrobial agent against *S. parasitica*. Linalool disrupted the cell membrane, resulting in the leakage of cellular components and restricting ribosomal function, thereby inhibiting protein synthesis. Comprehensive analyses revealed the specific role of linalool in protecting *S. parasitica* infection skin inflammation in grass carp. Meanwhile, the complement and coagulation systems, inflammatory factors, along with Actinobacteriota, significantly involved in this protective effect (**Figure 7**). Further investigation would be carried out to explore and refine how linalool affects the inflammatory effects of *S. parasitica* on fish through signaling pathways and mechanisms. We believe that this study will contribute to developing an innovative alternative strategy, distinct from antibiotics, to cope with *S. parasitica* infection in the aquaculture industry.

5. Materials and methods

5.1 Oomycete strains and materials

S. parasitica used in this study was from our lab, which was isolated from grass carp (*Ctenopharyngodon idella*). Linalool (0.862g/mL), pinene (0.874g/mL), limonene (0.842g/mL), myrcene ($\geq 95\%$), terpinene ($\geq 95\%$), resazurin dye, and Prodim iodide (PI) were purchased from Sigma-Aldrich (USA). SYTO 9 and FM4-64 were purchased from ThermoFisher (USA). Potato dextrose agar (PDA) and glucose yeast extract (GY) were purchased from Guangdong Huankai Microbial Sci. and Tech. Co., Ltd, China.

5.2 Agar diffusion assay

The anti-oomycetes effects of EOs and 5 components (linalool, terpinene, limonene, myrcene, and pinene) were assessed by using the agar diffusion method. Initially, a 6 mm mycelium plug was inoculated at the center of the plates. Subsequently, 6 mm sterile filter paper discs were soaked in EOs and components for more than 12 h and placed symmetrically at a distance of 2.5 cm from the target mycelium plug. Sterile distilled water was used as negative control. Finally, the width of the colonies was measured once the mycelium in the control group was completely covered the plate.

Inhibition rate (%) = $[(P1 - P2)/P1] \times 100\%$, where, P1 = colony growth diameter (mm) of control group, P2 = colony growth diameter (mm) of treated group.

5.3 Minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC) assays

To determine the MIC and MFC of linalool on mycelium, 6 mm mycelium plugs were inoculated into a 24 well cell culture plate. The wells contained linalool at concentrations of 0.8%, 0.4%, 0.2%, 0.1%, 0.05%, 0.025%, 0.0125%, and 0.00625%, dissolved in 1% Tween 20 and potato dextrose broth (PDB). Wells without mycelium plugs were served as negative controls. After incubating at 25°C for 48 h, the lowest concentration of linalool with no visible mycelium growth was recorded as MIC. Mycelium plugs showing no growth at this MIC concentration were then inoculated onto PDA plates and incubated for 7 days at 25°C to determine MFC.

The method for spores induction is similar as previously described by our previous research[27]. In a 96-well cell culture plate, 100 μ L of linalool at concentrations of 0.4%, 0.2%, 0.1%, 0.05%, 0.025%, 0.0125%, 0.00625%, and 0.003125% were dissolved in 1% Tween 20 and GY were added initially to each well, followed by the addition of 100 μ L of spore suspension at a

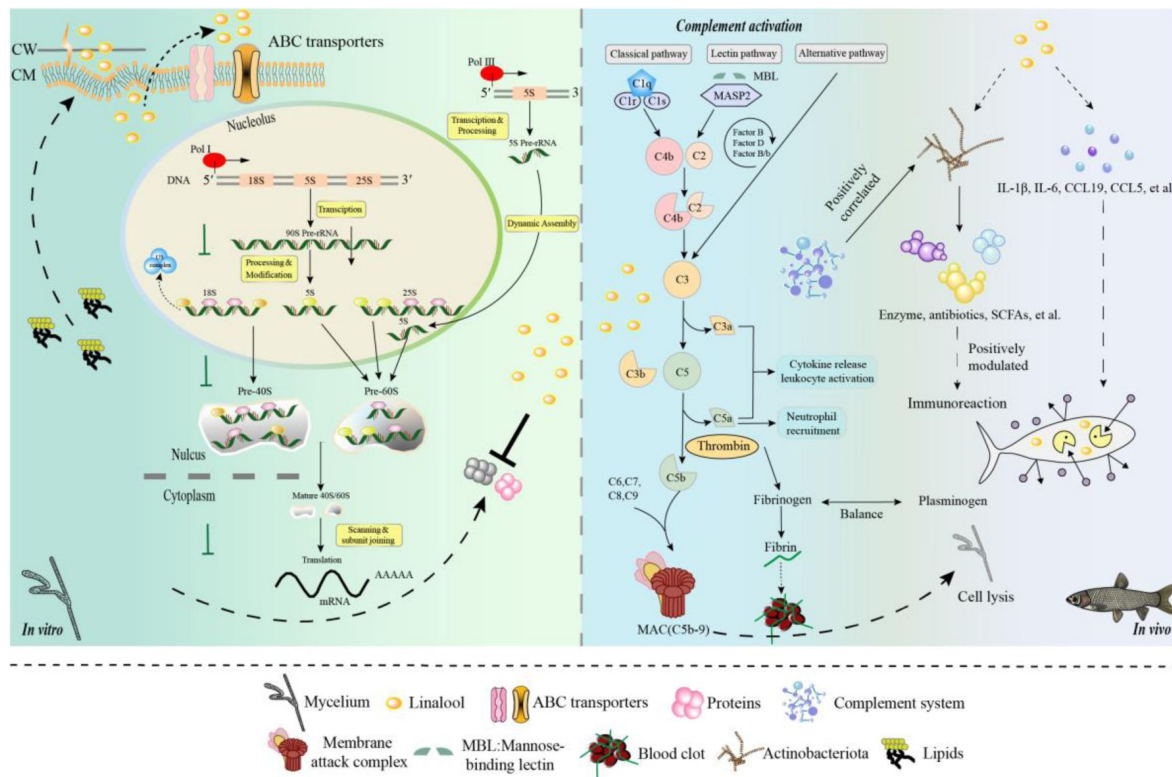


Figure 7.

Model diagram of the mode of action of linalool on *S. parasitica* and grass carp.

In vitro, (1) linalool influenced DNA transcription, tRNA transport, rRNA processing and maturation (5.8S, 18S, and 25S), and the biogenesis and assembly of ribosome subunits (40S and 60S) in the cell, which might lead to the reduction of *S. parasitica* growth; (2) Linalool disrupted the cell membrane, and the upregulation of glycerophospholipid metabolism likely represents the cell's response to cope with this damage; (3) ABC transporters contributed to metabolic resistance by pumping linalool out of the cell. *In vivo*, (1) Linalool enhanced the complement and coagulation system which in turn activated host immune defense and lysate *S. parasitica* cells; (2) Linalool promoted wound healing, tissue repair, and phagocytosis to cope with *S. parasitica* infection; (3) Linalool positively modulated the immune response by increasing the abundance of beneficial Actinobacteriota; (4) Linalool stimulated the production of inflammatory cytokines (*il-1 β* and *il-6*) and chemokines (*cc19* and *ccl5*) to lyse *S. parasitica* cells.

concentration of 1×10^6 spores/mL to each well. Malachite green and Tween 20 served as positive and negative controls, respectively. Following a 24-hour incubation at 20°C, 5 μ L of resazurin dye was added. When the color of the wells turned to pink, the linalool concentration in the first non-pink well corresponded to the MIC. The culture liquid from wells with no visible mycelium growth was streaked onto new plates. After incubating at 25°C for 7 days, the well with the lowest linalool concentration with no visible mycelium growth was determined as MFC.

5.4 Effect of linalool on the mycelium radial growth rate of *S. parasitica*

Different concentrations of linalool were added to molten PDA plates at 50°C for quantitative assessment. A similarly sized mycelium plug was placed at the center of each plate. Once the mycelium in the control group without linalool reached the edge of the plate (approximately 60 h), the radial diameter of the mycelium was measured. The calculation of growth inhibition was determined by assessing the percentage inhibition of radial growth, as below: $\text{PIRG \%} = [(R1 - R2)/R1] \times 100\%$, where, R1 = radial growth of control plate (mm), R2 = radial growth of treatment plate (mm)

5.5 Confocal laser scanning microscopy (CLSM) assay

5.5.1 Analysis of viability of *S. parasitica*

The mycelium was treated with linalool at MIC (0.05%) and 2×MIC (0.1%) for 2 h, and untreated control group was prepared simultaneously. The mycelium resuspended in PBS was further treated with PI at a final concentration of 2.5 g/mL for 15 minutes. PI is excited at a wavelength of 488 nm and emits light at 620 nm. Fluorescence images were observed using CLSM (LSM880, Zeiss, Germany) at 40× magnification.

5.5.2 Analysis of membrane damage of *S. parasitica*

Mycelium cultured for 18 h was collected and sequentially treated with SYTO 9 at a concentration of 20 μ mol/mL, FM4-64 at a concentration of 10 μ mol/mL, and linalool at a concentration of 3%. Record observations at 0, 2, 8, and 20 min. Both SYTO 9 and FM4-64 have an excitation wavelength of 488 nm and emit light at 620 nm and 520 nm, respectively. Fluorescence images were observed using CLSM (LSM880, Zeiss, Germany) at 40× magnification.

5.6 Scanning Electron Microscope (SEM) assay and transmission electron microscopy (TEM) assay

Mycelium exposed to MIC (0.05%) linalool for 2 h underwent fixation with 2.5% glutaraldehyde at 4°C for 4 h to examine the impact of linalool on membrane damage. Following a rinse with phosphate-buffered saline (PBS; pH 7.2), the fixed samples experienced dehydration for 15 min using a graded series of ethanol concentrations (10%, 30%, 50%, 70%, 90%, and 100%), followed by an additional 20 min in 100% ethanol. Post-dehydration, ethanol was exchanged twice for 20 min each with tertiary butanol. Subsequently, the samples were subjected to vacuum freeze-drying, gold-spraying, and examination with SEM (Hitachi, SU8010). After fixation, the samples were ultrathin-sectioned, subjected to negative staining using 1% phosphotungstic acid for 5 min and observed by TEM (Hitachi, HT-7700).

5.7 RNA-Seq and bioinformatics analysis

Total RNA was isolated from linalool treated (exposed to 0.5×MIC linalool for 10 minutes) and untreated control groups using the TRIzol method. The messenger RNA (mRNA) was enriched and fragmented into 200 bp fragments. These mRNA fragments were reverse-transcribed into cDNA

using random primers and reverse transcriptase. The cDNA ends were modified and ligated to an adaptor. After PCR amplification and purification, Illumina sequencing was performed. Quality control removed low-quality reads. The mapped reads were assembled using StringTie. Gene expression was quantified with the transcripts-per-million-reads method to identify differentially expressed genes (DEGs) using a threshold of $|\log \text{twofold change}| \geq 2$ and $\text{FDR} < 0.05$. Afterward, functional enrichment analysis was performed using the Kyoto Encyclopedia of Genes and Genomes (KEGG) database. KEGG pathway analysis was conducted using Goatools. The Majorbio cloud platform was used for data analysis[68].

5.8 Molecular docking

Linalool may interact with specific domains of the ribosome, potentially influencing its structure and function. Therefore, our investigation aimed to determine whether linalool could bind to the ribosome-associated upstream proteins of *S. parasitica*. As no crystal structure of *S. parasitica* ribosome proteins existed in the Protein Data Bank (PDB), we performed homology modeling. The protein sequences NOP1 (XP_012204452.1), SNU13 (XP_012198008.1), and DKC1 (XP_012193765.1) were obtained from GenBank, and homology modeling was conducted using AlphaFold2.

The most optimal configuration identified from the homologous modeling results was chosen as the receptor protein for subsequent molecular docking. The model was then evaluated using ProSA-web and PROCHECK server. The secondary structure of linalool was then loaded on PubChem. Both the ligand (linalool) and target proteins underwent processing with AutoDockTools 1.5.6, including energy minimization, dehydration, and hydrogenation. The conformation exhibiting the lowest energy was selected as the most likely binding state. The ultimate docking outcome was visualized utilizing the open-source software PyMOLv2 and Discovery Studio Visualizer V2021.

5.9 Animal Experiments

5.9.1 Ethics statement

All experiments involving animals were conducted according to the ethical policies and procedures approved by the Institutional Animal Care and Use Committee of Hunan Agricultural University, China (Approval No. 430516).

5.9.2 Linalool therapeutic and prophylactic assays

In this study, 4 groups were established: (1) Positive control (10 fish infected with *S. parasitica*). (2) Linalool therapeutic group (10 fish infected with *S. parasitica*, soaked in 0.00039% linalool in a 20L tank for 7 days). (3) Linalool prophylactic group (10 uninfected fish soaked in 0.00039% linalool in a 20L tank for 2 days, followed by the addition of 1×10^6 spores/mL secondary zoospores). (4) Negative control (10 uninfected fish without linalool treatment). Each group had 3 replicate tanks. The number of fish without visible mycelium on day 7 and the cumulative survival rate were recorded. The fish samples from the negative control, positive control, linalool therapeutic, and linalool prophylactic groups were collected for histopathological examinations. Tissue samples underwent fixation, sectioning, hematoxylin and eosin (H&E) staining, and microscopic examination using a microscope (BX53, Olympus).

5.9.3 16S rRNA and transcriptome sequencing and analysis

Genomic DNA was extracted from grass carp intestinal content samples and stored at -20°C following the instructions of the 16S rRNA gene pyrosequencing DNA kit. We conducted paired-end sequencing on the Illumina MiSeq platform, amplifying the V3-V4 regions with specific barcoded primers using a thermocycler PCR system. To unveil the intricate microbial diversity

within the samples, we employed Quantitative Insights Into Microbial Ecology (QIIME) which can compute and visualize α -diversity, β -diversity, and principal coordinate analysis (PCoA). Each sample was measured 3 times to ensure robust results.

Spleen tissue was obtained from 3 fish in each group on day 7. These collected spleen tissues were promptly flash-frozen in liquid nitrogen and then stored at -80°C until they were ready for RNA isolation. RNA-Seq and bioinformatics analysis was carried out as shown in [Section 2.6](#).

In each group, 8 fish were utilized for immunological assays, and on day 7, blood samples were collected from the tail veins using heparinized syringes and left to coagulate overnight at 4°C . After centrifugation at $3500\times g$ for 10 minutes at 4°C , serum was obtained from each fish and stored at -80°C for subsequent analysis. Kits from Nanjing Jiancheng Institute (Nanjing, China) were used to measure lysozyme (LZY) activity, superoxide dismutase (SOD) activity, and alkaline phosphatase (AKP) activity.

5.10 Detection of immune-related gene expression

Spleen tissues RNA extraction utilized the EZNA Total RNA Kit II (OMEGA R6934-02). RNA concentrations were quantified using a spectrophotometer (Eppendorf BioSpectrometer basic, Germany), and quality assessment was conducted through 1% gel electrophoresis. Subsequent cDNA synthesis employed reverse transcription with the RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, USA). RNA concentrations were quantified via spectrophotometry (Eppendorf BioSpectrometer basic, Germany). Subsequently, cDNA was synthesized via reverse transcription using the RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, USA). Expression levels of complement factor H (*cfh*), mannan-binding lectin serine protease 2 (*masp2*), complement component 1, r subcomponent (*c1r*), complement component 3 (*c3*), complement factor B (*cfb*), transcription factor AP-1 (*ap-1*), interleukin 1 beta (*il-1 β*) and interleukin 6 (*il-6*) were detected. The PCR procedure was described as follows: 95°C for 30 s, followed by 40 cycles of 95°C for 5 s and 60°C for 40 s.

Employing the CFX96 Touch Real-time PCR Detection System (Bio-rad, USA), gene expression levels were assessed using the $2^{-\Delta\Delta\text{Ct}}$ method, with β -actin serving as the reference gene. The primers in qPCR analysis are listed in [Table 1](#).

5.11 Statistical analysis

The data was expressed as mean $\pm$ standard error, and all experiments were conducted in triplicate. Pearson correlation analysis was employed to investigate the correlation between potential immune-related genes in grass carp and their characteristics, the method was implemented in R Programming language. Statistical analysis was performed using GraphPad Prism 9 software, and the student t-test was conducted.

Acknowledgements

This research was funded by National Natural Science Foundation of China (32073020, 32201960), Science and Technology Innovation Program of Hunan Province (2022RC1150), Hunan Provincial Natural Science Foundation of China (2023JJ40364), and Agricultural Science and Technology Innovation Fund of Hunan (2023CX49).

Primer name	Primer sequence 5'-3'
<i>c3</i> -F	ACAAGCCCATCTACACGCCT
<i>c3</i> -R	GAAACCTCAGGAATGGCGTA
<i>cfb</i> -F	AGAATGGAGAGGTGACCCCA
<i>cfb</i> -R	CCGCTATCCTGACACACTCG
<i>c1r</i> -F	GGGTTTCGGAGGGTTTGA
<i>c1r</i> -R	CCTGTATTGCCACATCTTTCA
<i>cfh</i> -F	ATAGGAGGGAAATGTGGGC
<i>cfh</i> -R	TCCCGTCCACGCCATC
<i>masp2</i> -F	TGGGCACCAGGTTCATCG
<i>masp2</i> -R	GCTGTTTGACTACGCCCATCTT
<i>il-1b</i> -F	ATCTCCTCGTCTGCTGGGTG
<i>il-1b</i> -R	TTGAGAGTTTCAGTGACCTCCTTC
<i>il-6</i> -F	AAAATCTGACCTCAACCCTGG
<i>il-6</i> -R	TGTCCACCCTTCCTCTTGCT
<i>ap-1</i> -F	TCTCACTCAACAGATGCCAAAC
<i>ap-1</i> -R	CTTAGCGGCTGCGAGTTTAT
β -ActinF	GCTATGTGGCTCTTGACTTCG
β -ActinR	GGGCACCTGAACCTCTCATT

Table 1.

Primers for qPCR were used in this study.

Author contributions

Tao Tang was in charge of conceptualization, methodology, writing-original draft, writing-review & editing.

Weiming Zhong was in charge of data curation.

Puyu Tang was in charge of methodology.

Rongsi Dai was in charge of data curation.

Jiajing Guo was in charge of writing - review & editing.

Zhipeng Gao was in charge of funding acquisition, supervision, writing-original draft, writing-review & editing.

Declaration of interests

The authors declare no competing interests.

Data and code availability

The data that support the findings of this study are available in the supplementary material of this article and are available from the corresponding author upon reasonable request.

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Reviewer #1 (Public review):

Summary:

The work seeks to investigate the efficacy of linalool as a natural alternative for combating *Saprolegnia parasitica* infections, which would provide great benefit to aquaculture. This paper shows the effect of linalool in vitro using a variety of techniques including changes in *S. parasitica* membrane integrity following linalool exposure and alterations in cell metabolism and ribosome function. Additionally, this work goes on to show that prophylactic and concurrent treatment of linalool at the time of *S. parasitica* infection can improve survival and tissue damage in vivo in their grass carp infection model. The conclusions of the paper are partially supported by the data, cleaning up, clarifying, and elaborating on some aspects of this work is necessary.

(1) Adding microscopy of the untreated group to compare Figure 2A with would further strengthen the findings here.

(2) Quantification of immune infiltration and histological scoring of kidney, liver, and spleen in the various treatment groups would increase the impact of Figure 4.

(3) The data in Figure 6 I is not sufficiently convincing as being significant.

(4) Comparisons of the global transcriptomic analysis of the untreated group to the PC, LP, and LT groups would strengthen the author's claims about the immunological and transcriptomic changes caused by linalool and provide a true baseline.

<https://doi.org/10.7554/eLife.100393.1.sa2>

Reviewer #2 (Public review):**Summary:**

In this study, the authors aimed to delineate the antimicrobial activity of linalool and tried to investigate the mode of action of linalool against *S. parasitica* infection. One of the main focuses of this work was to identify the in vitro and in vivo mechanisms associated with the protective role of linalool against *S. parasitica* infection.

Strengths:

- (1) The authors have used a variety of techniques to prove their hypothesis.
- (2) An adequate number of replicates were used in their studies.
- (3) Their findings showed a protective role of linalool against oomycetes and makes it an attractive future antibiotic in the aquaculture industry.

Weaknesses:

There are several weaknesses in this manuscript.

- (1) The authors have taken for granted that the readers already know the experiments/assays used in the manuscript. There was not enough explanation for the figures as well as figure legends.
- (2) The authors missed adding the serial numbers to the references.
- (3) The introduction section does not provide adequate rationale for their work, rather it is focused more on the assays done.
- (4) Full forms are missing in many places (both in the text and figure legends), also the resolution of the figures is not good. In some figures, the font size is too small.
- (5) There is much mislabeling of the figure panels in the main text. A detailed explanation of why and how they did the experiments and how the results were interpreted is missing.
- (6) There is not enough experimental data to support their hypothesis on the mechanism of action of linalool. Most of the data comes from pathway analysis, and experimental validation is missing.

Overall, the conclusions drawn by the authors are partially justified by the data. Importantly, this paper has discovered the novelty of the compound linalool as a potent antimicrobial agent and might open up future possibilities to use this compound in the aquaculture industry.

<https://doi.org/10.7554/eLife.100393.1.sa1>

Author response:**Reviewer #1:**

(1) Adding microscopy of the untreated group to compare Figure 2A with would further strengthen the findings here.

Thank you for your comments on our manuscript. We will carefully revise this part. Actually, we used a time-lapse method to capture images at 0 minute before any drugs were added. We

will change '0 min' to 'untreated,' which will further strengthen our findings.

(2) Quantification of immune infiltration and histological scoring of kidney, liver, and spleen in the various treatment groups would increase the impact of Figure 4.

Thank you for your comments on our manuscript. To further strengthen Figure 4, we will use quantification of immune infiltration and histological scoring of the kidney, liver, and spleen in different groups. Additionally, we will use ImageJ software for molecular immunohistochemistry and determine the ratio of normal to abnormal cells, providing more comprehensive insights into the effects of the treatments.

(3) The data in Figure 6 I is not sufficiently convincing as being significant.

Thank you for your comments on our manuscript. Previous researches have shown that antibiotics and other drugs can cause alterations in gut microbiota. Therefore, we plan to study the effects of linalool on gut microbiota. The results of this part were mostly built on gut microbiota sequencing and correlation analysis, we have tried several times to isolate vital microbes from the gut, but this is a very challenging work and the results were not good. Thus, in this study, we just predicted the effects of linalool on gut microbiota. In the future, we will continue to delve into interesting aspects of how linalool affects gut microbiota.

(4) Comparisons of the global transcriptomic analysis of the untreated group to the PC, LP, and LT groups would strengthen the author's claims about the immunological and transcriptomic changes caused by linalool and provide a true baseline.

Thank you for your comments on our manuscript. We will compare the global transcriptomic analysis of the untreated group with the PC, LP, and LT groups to strengthen the claims about the immunological and transcriptomic changes induced by linalool, thereby providing a true baseline.

Reviewer #2:

(1) The authors have taken for granted that the readers already know the experiments/assays used in the manuscript. There was not enough explanation for the figures as well as figure legends.

Thank you for your comments on our manuscript. We will provide more detailed explanations of the experiments and assays used in the manuscript, as well as enhance the descriptions in the figure legends, to ensure that readers have a clear understanding of the figures and context.

(2) The authors missed adding the serial numbers to the references.

Thank you for your comments on our manuscript. We will add serial numbers to the references to ensure proper citation and improve the clarity of our manuscript.

(3) The introduction section does not provide adequate rationale for their work, rather it is focused more on the assays done.

Thank you for your comments on our manuscript. We will add a section to the introduction that provides a rationale for our work, specifically focusing on the impact of plant extract on immunoregulation.

(4) Full forms are missing in many places (both in the text and figure legends), also the resolution of the figures is not good. In some figures, the font size is too small.

Thank you for your comments on our manuscript. We will ensure that all abbreviations are expanded where necessary, both in the text and figure legends. Additionally, we will improve the resolution of the figures and increase the font size where needed to enhance clarity.

(5) There is much mislabeling of the figure panels in the main text. A detailed explanation of why and how they did the experiments and how the results were interpreted is missing.

Thank you for your comments on our manuscript. We will improve the labeling of the figure panels, provide detailed explanations of the experimental methods, including their rationale and interpretation, and clarify the connections between the methods.

(6) There is not enough experimental data to support their hypothesis on the mechanism of action of linalool. Most of the data comes from pathway analysis, and experimental validation is missing.

Thank you for your comments on our manuscript. We have tried our best to link transcriptomic data, pathway analysis, experimental validation. We carried out many experiments to substantiate the changes inferred from the transcriptomic data as SEM, TEM, CLSM, molecular docking, RT-qPCR, histopathological examinations. The detailed information is listed as follows. (1) As shown in Figure 2, we combined the transcriptomic data related to membrane and organelle with SEM, TEM, and CLSM images. After deep analysis of these data and observation together, we illustrated that cell membrane may be a potential target for linalool. (2) As shown in Figure 3, we carried out molecular docking to explore the specific binding protein of linalool with ribosome which were screen out as potential target of linalool by transcriptomic data. (3) As shown in Figure 5, transcriptomic data illustrated that linalool enhanced the host complement and coagulation system. To substantiate these changes, we carried out RT-qPCR to detect those important immune-related gene expressions, and found that RT-qPCR analysis results were consistent with the expression trend of transcriptome analysis genes. (4) As shown in Figure 4 and 5, transcriptomics data revealed that linalool promoted wound healing tissue repair, and phagocytosis (Figure. 5E). To ensure these, we carried out histopathological examinations, and found that linalool alleviated tissue damage caused by *S. parasitica* infection on the dorsal surface of grass carp and enhancing the healing capacity (Figure. 4G). But we know the antimicrobial mechanism of linalool need further investigation, we will conduct more experiments to explore the antimicrobial mechanism of action of linalool in the future.

<https://doi.org/10.7554/eLife.100393.1.sa0>